Tanishk Jain

Noida, India | 9315747674 | tanishkjain3011@gmail.com | LinkedIn | GitHub

B.Tech Computer Science (AI & ML) student with experience across machine learning, computer vision, and full stack development. Built end-to-end deep learning pipelines for medical image segmentation and deployed production web applications.

Education

VIT Bhopal University 2023 – 2027
B.Tech in Computer Science Engineering (AI & ML), CGPA: 8.34 / 10.0 *Bhopal, India*

Experience

Indian Institute of Technology Delhi June 2026 – September 2026
Machine Learning Research Intern *Delhi, India*

- Developed deep learning pipelines for retinal vessel segmentation and artery/vein classification under Prof. Tapan Kumar Gandhi.
- Trained and benchmarked encoder–decoder segmentation architectures including Attention U-Net, tuning loss functions and hyperparameters against IoU and Dice coefficient.

DJUBO Jan 2026 – Apr 2026
Software Developer Intern | Repository *Noida, India*

- Built a large-scale employee attendance management platform using AngularJS, Django REST Framework, SQLite, and token-based authentication.
- Implemented REST endpoints powering dashboards, calendar views, admin workflows, leave management, and analytics visualizations via Handsontable and Highcharts.

Projects

Parkinson's Detection from Voice with Explainable AI | *Python, Scikit-Learn, XGBoost, SHAP, LIME* GitHub

- Identified speaker leakage in conventional train-test splits, showing models were learning speaker identity rather than disease characteristics.
- Evaluated 12 models across 960 experimental runs, showing conventional evaluation inflated ROC-AUC by 0.159 on average and changed model rankings.
- Applied SHAP and LIME on unseen speakers, achieving 0.924 ROC-AUC with a reduced two-feature screening model at 92% sensitivity.

Real-Time Attention Monitoring System | *Python, OpenCV, MediaPipe* GitHub

- Built a real-time webcam system that analyzes facial landmarks, head pose, and posture to detect distraction.
- Combined these signals in a rule-based scoring engine to monitor attention frame by frame.

Hisaab: Group Trip Expense Manager | *React, TypeScript, Supabase, Tailwind CSS* GitHub | Live

- Built and deployed a group expense manager supporting optimized settlements, uneven splits, and multi-payer transactions.
- Implemented offline-first storage with background Postgres sync and row-level security.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript
AI / ML: TensorFlow, Scikit-Learn, OpenCV, MediaPipe; Deep Learning, Computer Vision, Image Segmentation
Libraries: NumPy, Pandas, Matplotlib, SHAP, LIME
Full Stack: React, AngularJS, Django REST Framework, Supabase, SQL
Tools: Git, GitHub, Jupyter, Google Colab

Leadership & Achievements

Microsoft Technical Club, VIT Bhopal Dec 2024 – May 2026
Vice President *Bhopal, India*

- Led planning and delivery of SIH Success Blueprint, coordinating Smart India Hackathon finalists, speaker onboarding, and promotions.

VIT Bhopal x Johns Hopkins University HealthHack February 2026
Team Leader, Finalist | Repository *Bhopal, India*

- Led a team of 6 building CareFast, a mobile-first emergency healthcare application, advancing to the Final Round.

Certifications

Applied Machine Learning, University of Michigan | AWS Technical Essentials | ServiceNow Certified System Administrator (CSA)